

MATH 101-
Mathematical Analysis 1

**ANTI-DERIVATIVES AND THE INDEFINITE
INTEGRAL
(Basic Rules)**

MS. MAYCA N. ODEÑA
Instructor

DEFINITION

- ▶ The process of finding the Anti-Derivatives of a given function is called *antidifferentiation* or *integration*.
- ▶ An anti-derivative, also known as an *Indefinite Integral*, is a function F whose derivative is a given function f within a specific interval I , such that $F'(x) = f(x)$ for every value of x in that interval I .

DEFINITION

- Integration actually reverses what differentiation has done. Since $f(x)$ is the derivative of $F(x)$, then by substituting $F'(x)$ for $f(x)$, we have:

$$\int F'(x) dx = F(x) + C$$

FINDING THE ANTIDERIVATIVE S OF THE GIVEN FUNCTIONS

1. The Constant Rule

$$\int \mathbf{0} \, dx = C$$

$$\int \mathbf{k} \, dx = kx + C$$

EXAMPLES:

$$\int \mathbf{3} \, dx = 3x + C$$

$$\int \mathbf{2} \, dx = 2x + C$$

$$\int \mathbf{99} \, dx = 99x + C$$

2. The Constant Multiple Rule

$$\int \mathbf{k} \bullet \mathbf{f(x)} \, \mathbf{dx} = \mathbf{k} \int \mathbf{f(x)} \, \mathbf{dx} + \mathbf{C}$$

EXAMPLE:

$$\begin{aligned} \int \mathbf{5x} \, \mathbf{dx} &= \mathbf{5} \int \mathbf{(x)} \, \mathbf{dx} \\ &= \mathbf{5} \bullet \frac{x^2}{2} + \mathbf{C} \\ &= \frac{5}{2} x^2 + \mathbf{C} \end{aligned}$$

3. The Power Rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad \text{where } n \neq -1$$

EXAMPLES:

$$\begin{aligned} \int x^3 dx &= \frac{x^{3+1}}{3+1} + C \\ &= \frac{x^4}{4} + C \\ &\text{or } \frac{1}{4}x^4 + C \end{aligned} \qquad \begin{aligned} \int 5x^3 dx &= 5 \cdot \frac{x^{3+1}}{3+1} + C \\ &= \frac{5x^4}{4} + C \\ &\text{or } \frac{5}{4}x^4 + C \end{aligned}$$

4. The Sum and Difference Rule

$$\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$$

EXAMPLES:

$$\begin{aligned}\int (x + 4) dx &= \int (x) dx + \int (4) dx \\ &= \frac{x^{1+1}}{1+1} + C_1 + 4x + C_2 \\ &= \frac{x^2}{2} + 4x + C \text{ or } \frac{1}{2}x^2 + 4x + C\end{aligned}$$

EXAMPLES:

$$\int (2x^3 + 4x^2 - x) dx = 2 \int (x^3) dx + 4 \int (x^2) dx - \int (x) dx$$

$$= 2 \left[\frac{x^{3+1}}{3+1} \right] + C_1 + 4 \left[\frac{x^{2+1}}{2+1} \right] + C_2 - \left[\frac{x^{1+1}}{1+1} \right] + C_3$$

$$= 2 \left[\frac{x^4}{4} \right] + 4 \left[\frac{x^3}{3} \right] - \left[\frac{x^2}{2} \right] + C$$

$$= \frac{x^4}{2} + \frac{4x^3}{3} - \frac{x^2}{2} + C \text{ or } \frac{1}{2} x^4 + \frac{4}{3} x^3 - \frac{1}{2} x^2 + C$$

5. Exponential Rule

$$\int e^{kx} dx = \frac{1}{k} e^{kx} + C \text{ where } k \neq 0$$

$$\int e^x dx = e^x + C$$

EXAMPLES:

$$\begin{aligned}\int 7e^x dx &= 7 \int e^x dx \\ &= 7 (e^x) + C \\ &= 7e^x + C\end{aligned}$$

$$\begin{aligned}\int 2e^{3x} dx &= 2 \int e^{3x} dx \\ &= 2 \cdot \frac{1}{3} e^{3x} + C \\ &= \frac{2}{3} e^{3x} + C\end{aligned}$$

6. Trigonometric Function Rule

$$\int \sin x \, dx = -\cos x + C$$

$$\int \csc^2 x \, dx = -\cot x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sec x \tan x \, dx = \sec x + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \csc x \cot x \, dx = -\csc x + C$$

EXAMPLES:

$$\int (3\cos x - 2\sec^2 x + 5\sin x) dx$$

$$= 3 \int \cos x dx - 2 \int \sec^2 x dx + 5 \int \sin x dx$$

$$= 3\sin x - 2\tan x + 5(-\cos x) + C$$

$$= 3\sin x - 2\tan x - 5\cos x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C$$

EXAMPLES:

$$\int (4\csc x \cot x + \frac{1}{2} \sec x \tan x - 6\csc^2 x) dx$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$= 4 \int \csc x \cot x dx + \frac{1}{2} \int \sec x \tan x dx - 6 \int \csc^2 x dx$$

$$= 4(-\csc x) + \frac{1}{2} \sec x - 6(-\cot x) + C$$

$$= -4\csc x + \frac{1}{2} \sec x + 6\cot x + C$$

**Thank you for
listening!**

